Homothety and centres of similitude

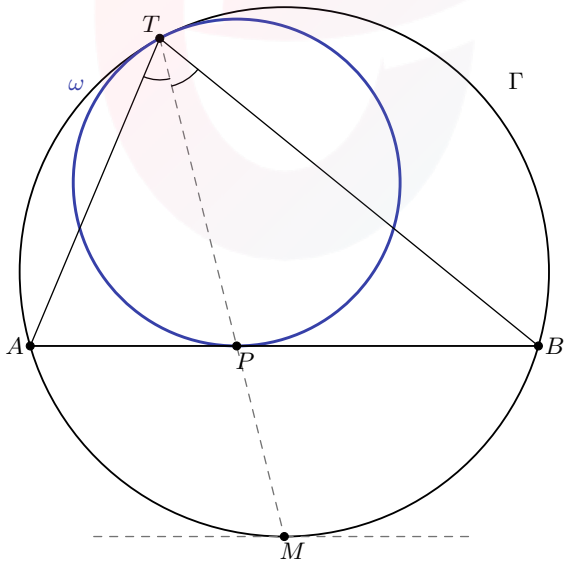
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L4-X6 Level 4 Olympiad · Geometry · Beyond the textbook

Homothety as a proof tool, covering centres of similitude, touching circles, Monge's theorem, the incircle and excircle lemma and the nine-point circle.
By the end you can find the homothety in a figure with touching circles, parallel lines or circles in an angle; use centres of similitude and Monge's theorem to prove collinearity and concurrency; use the incircle and excircle lemma and the nine-point circle as homothetic images.
Prerequisites: Olympiad angle chasing (L4-GE1), Lengths and ratios (L4-GE2). **Contests:** BMO1, MOG, AIME, USAJMO. **Time:** about 120 minutes.

1 Lesson

Circle ω lies inside circle Γ (centre O) and touches Γ at T. A chord AB of Γ touches ω at P. Prove that TP bisects $\angle ATB$. Enlarge ω from T into Γ. The scale factor exceeds 1, so the tangent AB goes to a parallel tangent of Γ farther from T, and P goes to its contact point M. So T, P, M are collinear, and $OM \perp AB$ makes M the midpoint of arc AB not containing T.



This is GCSE enlargement, called *homothety* in olympiads. Textbooks redo each such fact with fresh similar triangles, but one map does the work of all of them. The three-circle theorem is Monge's, and Euler found the line OGH in 1765.

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1.1 Homothety

Theorem 1.
The homothety with centre S and ratio $k \neq 0$ sends X to X' with $\overrightarrow{SX'} = k\overrightarrow{SX}$.
(1) $\overrightarrow{X'Y'} = k\overrightarrow{XY}$, so lines go to parallel lines, lengths scale by $|k|$, angles are kept.
(2) The circle with centre P and radius r goes to the circle with centre P' and radius $|k|r$, and tangency is kept.
(3) Ratio k_1 at S_1 followed by ratio k_2 at S_2, with $k_1k_2 \neq 1$, is a homothety with ratio k_1k_2 and centre on line S_1S_2.

Proof. (1) $\overrightarrow{X'Y'} = \overrightarrow{SY'} - \overrightarrow{SX'}$, and (2) follows. (3) The composite equals $X \mapsto Z + k_1k_2(X - Z)$ for

$$Z = \frac{(1-k_2)S_2 + k_2(1-k_1)S_1}{1-k_1k_2},$$

whose weights add up to 1.

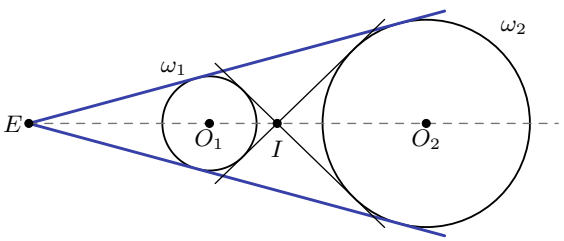
Theorem 2 (spotting a homothety).
If $\overrightarrow{A'B'} = k\overrightarrow{AB}$ with $k \neq 1$, a homothety with ratio k sends $A \mapsto A'$, $B \mapsto B'$, so AA' and BB' pass through its centre. Hence if triangles ABC, $A'B'C'$ have corresponding sides parallel and are not translates, AA', BB', CC' are concurrent.

Proof. Take $S = \frac{A'-kA}{1-k}$. Then $S + k(B-S) = A' + k(B-A) = B'$. For triangles, $k\overrightarrow{AB} + k'\overrightarrow{BC} \parallel \overrightarrow{AB} + \overrightarrow{BC}$ forces $k = k'$.

1.2 Two circles

Theorem 3 (centres of similitude).
Let ω_1, ω_2 have centres $O_1 \neq O_2$ and radii r_1, r_2. The homotheties taking ω_1 to ω_2 have ratios r_2/r_1 (if $r_1 \neq r_2$) and $-r_2/r_1$, and their centres E and I are the *external* and *internal centres*, divide O_1O_2 externally and internally in the ratio $r_1 : r_2$.
(1) Common external tangents pass through E, internal ones through I.
(2) If the circles touch at T, then $T = I$ for external touching and $T = E$ for internal.

Proof. The map sends $O_1 \mapsto O_2$ and scales radii by $|k|$, so $k = \pm r_2/r_1$ and $\overrightarrow{SO_2} = k\overrightarrow{SO_1}$ fixes S. (1) For a common tangent touching at T_1, T_2, $\overrightarrow{O_2T_2} = \pm\frac{r_2}{r_1}\overrightarrow{O_1T_1}$ (+ if the circles are on the same side), so apply Theorem 2. (2) T is on O_1O_2 with $TO_1 = r_1$, $TO_2 = r_2$.



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Theorem 4 (Monge).
For three circles with distinct centres and pairwise different radii, the three external centres are collinear, and the internal centres of two pairs are collinear with the external centre of the third.

Proof. The homothety at E_{12} then at E_{23} takes ω_1 to ω_3 with ratio $\frac{r_3}{r_1} \neq 1$. By Theorem 1(3) the centre of this map is on $E_{12}E_{23}$, and by Theorem 3 the centre is E_{13}. With I_{12}, I_{23} the ratio is $\left(-\frac{r_2}{r_1}\right)\left(-\frac{r_3}{r_2}\right) = \frac{r_3}{r_1}$ again.

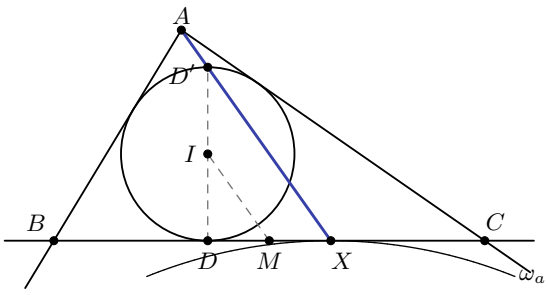
Remark (Desargues). If triangles ABC and $A'B'C'$ have $AB \parallel A'B'$, $BC \parallel B'C'$ and $CA \parallel C'A'$, a homothety or a translation takes one to the other, so AA', BB', CC' are concurrent or parallel. This is Desargues' theorem with the axis of perspective at infinity. The general theorem, two triangles perspective from a point are perspective from a line, is the projective form of the same statement, and Monge's theorem follows from Desargues in the same way.

Remark (Miquel). A spiral similarity is a homothety followed by a rotation about the same centre. The Miquel point of a complete quadrilateral is the centre of the spiral similarity taking one pair of opposite sides to the other, so the arguments of this unit carry over to L5-GE2.

1.3 The incircle and the excircle

Lemma 5.
The incircle of ABC (centre I, radius r) touches BC at D, and DD' is a diameter. The A-excircle (centre I_a, radius r_a) touches BC at X. Then
(1) A, D', X are collinear and $AD' : AX = r : r_a$,
(2) $BX = CD = s - c$, so D, X are symmetric about the midpoint M of BC, and
(3) $MI \parallel AX$.

Proof. The homothety h at A with ratio r_a/r fixes lines AB, AC, so h takes the incircle to the excircle. With n the unit normal to BC towards A, $D' = I + rn$ and $X = I_a + r_a n = h(D')$. For (2), if the excircle touches AB, AC at Z, Y, then $AY + AZ = (c + BX) + (b + CX) = 2s$, so $AZ = s$ and $BX = BZ = s - c = CD$. (3) is the midline of triangle $DD'X$.



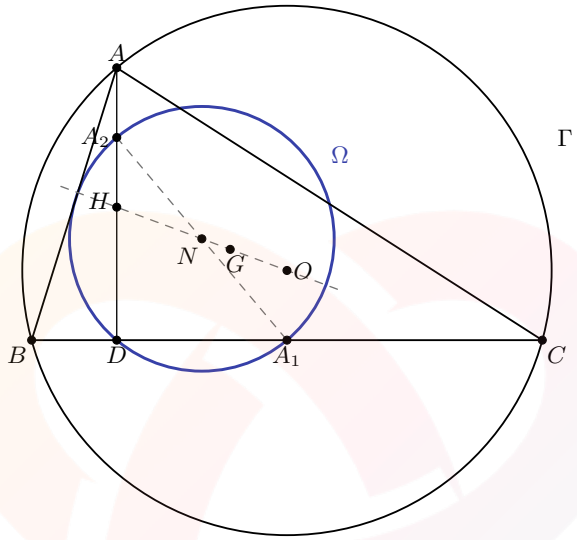
1.4 The nine-point circle

Theorem 6.
Let ABC have circumcircle Γ (centre O, radius R), centroid G, orthocentre H, and let Ω be the image of Γ under the homothety at H with ratio $\frac{1}{2}$.
(1) Ω passes through the midpoints of the sides, the feet of the altitudes and the midpoints

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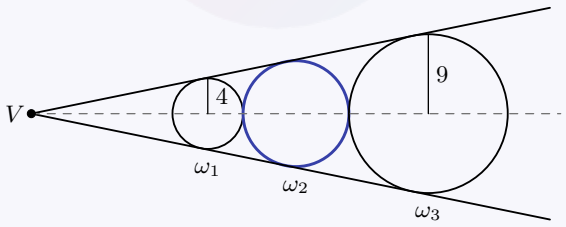
of AH, BH, CH, and its centre N is the midpoint of OH.
(2) The homothety at G with ratio $-\frac{1}{2}$ also takes Γ to Ω, sending A, B, C to the midpoints of the opposite sides and O to N.
(3) H, G, N, O are collinear (the Euler line), with $HG = 2\,GO$.

Proof. With origin O, $h = a + b + c$ satisfies $(h - a) \cdot (b - c) = |b|^2 - |c|^2 = 0$, so $h = H$, and $G = h/3$ gives (3). The map $X \mapsto \frac{X+h}{2}$ sends a to the midpoint A_2 of AH and $-a$ to the midpoint A_1 of BC, so A_1A_2 is a diameter of Ω, and $\angle A_2DA_1 = 90^\circ$ puts the foot D on Ω. The map $X \mapsto \frac{h-X}{2}$ sends $a \mapsto A_1$ and $0 \mapsto N$. ■



1.5 Worked examples

Example 1.
Circles $\omega_1, \omega_2, \omega_3$ lie inside an angle with vertex V, each touching both arms, and ω_2 touches ω_1 and ω_3, and ω_1 is nearest to V. The radii of ω_1 and ω_3 are 4 and 9. Find the radius of ω_2.



Solution.
The standard way is to take half-angle θ and radius x, the tangencies give

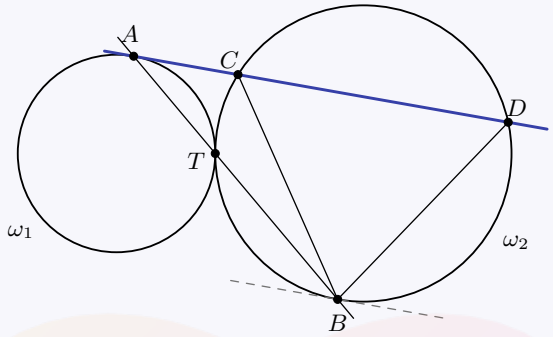
$$\frac{x-4}{\sin\theta} = x + 4, \qquad \frac{9-x}{\sin\theta} = 9 + x.$$

Dividing, $2x^2 = 72$ and $x = 6$.
With a homothety, note that V is the external centre of each pair. The homothety at V with ratio $x/4$ takes ω_1 to ω_2, hence ω_2 to the inscribed circle touching ω_2 beyond ω_2, which is ω_3. So $9 = x \cdot \frac{x}{4}$ and $x = 6$.

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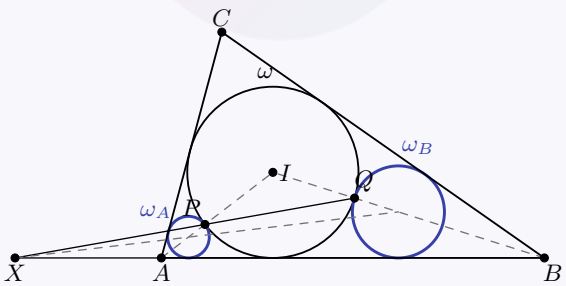
Remark. Radii of a chain of circles in an angle form a geometric progression, and the angle never enters.

Example 2.
Circles ω_1 and ω_2 touch externally at T. A line through T meets ω_1 again at A and ω_2 again at B. The tangent to ω_1 at A meets ω_2 at C and D. Prove that $BC = BD$.



Solution.
The homothety at T taking ω_1 to ω_2 sends A to the other point of ω_2 on line TA, which is B, and the tangent at A to the tangent at B. So CD is parallel to the tangent at B, hence perpendicular to O_2B, where O_2 is the centre of ω_2. Thus O_2B is the perpendicular bisector of chord CD, and $BC = BD$. ■
Remark. With touching circles, try the homothety at the contact point first.

Example 3.
Triangle ABC has incircle ω with centre I and radius r, and $CA \neq CB$. The circle ω_A inside the triangle touches AB, AC and touches ω externally, and ω_B is defined in the same way at B. Segments AI and BI meet ω at P and Q. Prove that line PQ, line AB and the line through the centres of ω_A and ω_B are concurrent.



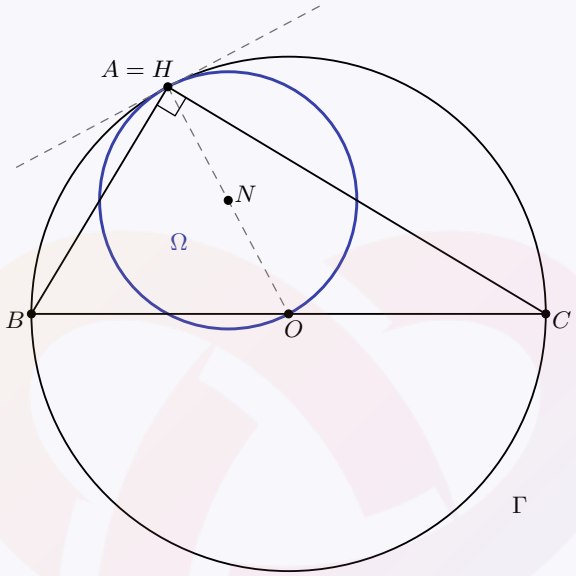
Solution.
By Theorem 3, A and P are the external and internal centres of ω_A and ω. Similarly B and Q for ω_B and ω.
As $\angle A \neq \angle B$, the radii differ, since the homothety at A taking ω to ω_A sends the far end of the diameter of ω on AI to P, so

$$\frac{\rho_A}{r} = \frac{AI - r}{AI + r} = \frac{1 - \sin\frac{A}{2}}{1 + \sin\frac{A}{2}},$$

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which decreases in A.
So the external centre X of ω_A and ω_B, which is on their line of centres, lies on AB and on PQ by Monge's theorem for ω_A, ω_B, ω. ■
Remark. Menelaus would need the radii exactly, but here we only needed them unequal.

Example 4.
Prove that the nine-point circle of a triangle touches its circumcircle if and only if the triangle has a right angle.



Solution.
If the triangle is equilateral the circles are concentric. Otherwise, if Γ and Ω touch at T, then T is a centre of similitude, H or G. As G is inside Γ, $T = H$ lies on Γ.
With origin O, $|a+b+c|^2 = R^2$ gives $a\cdot b + b\cdot c + c\cdot a = -R^2$, so $(a+b)\cdot(a+c) = 0$, that is $OC_1 \perp OB_1$ for the midpoints C_1, B_1 of AB, AC. If $C_1 = O$ or $B_1 = O$, a side is a diameter, and otherwise $AB \perp AC$.
Conversely, if $\angle A = 90^\circ$ then $H = A$ lies on Γ and on Ω, and the homothety at H sends the tangent to Γ at H to a parallel tangent to Ω through H, which is the same line. ■

Common traps.

- Circles touching externally have the contact point as internal centre, ratio negative.
- Equal radii give no external centre (a translation), so check before using Monge.
- Two pairs of parallel sides do not make triangles homothetic.
- In a proof, state centre, ratio and what goes to what, because a line may meet the image circle twice, so justify each image point.

1.6 How to spot a homothety

1. Touching circles suggest the contact point.

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2. Parallel segments, or a tangent parallel to a chord, suggest Theorem 2.
3. Circles in one angle suggest the vertex, and a diameter of the incircle suggests Lemma 5.
4. Midpoints of sides or of AH, BH, CH suggest Theorem 6.
5. Three circles suggest Monge.

2 Practice problems

The problems are roughly in order of difficulty, from $\star$ to $\star\star\star\star$. Hints are in Section 4 and full solutions in Section 5.

Problem 1 $\star$ Three coins of radii 1, 2 and 3 lie flat on a table, each touching the other two. Line ℓ passes through the point where the coins of radii 1 and 2 touch and the point where the coins of radii 2 and 3 touch. Find the distance from the centre of the coin of radius 1 to the point where ℓ meets the line through the centres of the coins of radii 1 and 3.

Problem 2 $\star\star$ The incircle of triangle ABC touches BC at D, and DD' is a diameter of the incircle. Line AD' meets BC at X. Prove that D' is the midpoint of AX if and only if $AB + AC = 3\,BC$.

Problem 3 $\star\star$ Circle ω lies inside circle Γ and touches Γ at T. A chord AB of Γ touches ω at P, and M is the midpoint of the arc AB of Γ not containing T. Given $MA = 6$ and $MP = 4$, find PT.

Problem 4 $\star\star\star$ Circles ω_1 and ω_2 lie outside each other. A common external tangent touches them at T_1 and T_2 and meets their line of centres at E. A line through E meets ω_1 at A and B and ω_2 at C and D, in the order E, A, B, C, D. Prove that $EA \cdot ED = ET_1 \cdot ET_2$.

Problem 5 $\star\star\star$ Squares are drawn outwards on the three sides of a right-angled triangle with sides 3, 4 and 5. For each square, the side opposite the triangle is extended to a full line. Find the area of the triangle bounded by these three lines.

Problem 6 $\star\star\star\star$ Let ABC be a triangle that is not equilateral, with incentre I, circumcentre O and excentres I_a, I_b, I_c. The incircle touches BC, CA, AB at D, E, F. Prove that the lines I_aD, I_bE, I_cF are concurrent at a point of line OI.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **BMO1 2012/13 Problem 2** $\star\star$
   Look at the homothety at X taking S to T and see where the homothety sends A.
2. **IMO 1978 Problem 4** $\star\star$
   Compare the circle with the circumcircle via the homothety at their tangency point, and watch where the arc midpoint goes.
3. **IMO 1981 Problem 5** $\star\star$
   Locate the centres of the three circles on the angle bisectors and compare their triangle with ABC.

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4. **BMO1 2011/12 Problem 3** ⋆⋆⋆
   Use the homothety at X taking S to S1 and the one at Y taking S to S2. Where do P-related points go?

5. **AIME 2007 II Problem 15** ⋆⋆⋆
   The centres of the three equal circles and the incentre are related by a homothety, so find its ratio and then use the incircle radius.

6. **Baltic Way 2011 Problem 14 (via imogeometry.blogspot.com Baltic Way index)** ⋆⋆⋆
   Use the diameter FG of the incircle and the homothety at C (or the excircle one) that carries the incircle to a circle tangent to AB.

7. **Baltic Way 2007 Problem 15** ⋆⋆⋆
   Look at the homothety at B between the incircle and the second circle, and where D goes.

8. **Baltic Way 1992 Problem 19** ⋆⋆⋆
   Send each small circle to C by its homothety at A or B and track the tangent line t.

9. **BMO2 2015/16 Problem 1** ⋆⋆⋆
   For two touching circles and a common tangent, express the distance between the touch points in terms of the radii.

10. **AIME I 2019 Problem 15** ⋆⋆⋆⋆
    Homothety at each tangency point sends the small circle to omega, and combine with the radical axis PQ and power of P.

11. **IMO Shortlist 1999 G6** ⋆⋆⋆⋆
    Use the homothety at M taking $Omega_1$ to Omega and see what happens to the chord CD and to the line AB.

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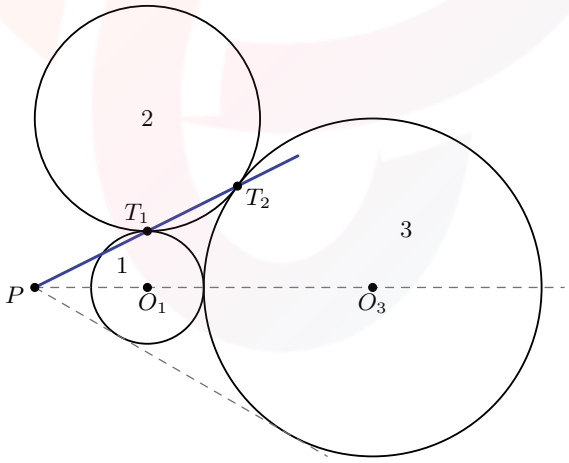
4 Hints

1. Each point of contact is a centre of similitude of its two coins. Which kind?
2. Lemma 5 gives $AD' : AX$. Write both radii in terms of the area.
3. First show that T, P, M are collinear. Then look for two similar triangles with a common vertex M.
4. Which homothety sends A to C? Then use the power of E with respect to ω_1.
5. The new triangle has its sides parallel to those of the old one. Each side moves out by its own length, so where must the centre of the homothety be?
6. Compare EF with line I_bI_c. Then find the circumcentre of triangle $I_aI_bI_c$ using Theorem 6.

5 Solutions

Problem 1

Three coins of radii 1, 2 *and* 3 *lie flat on a table, each touching the other two. Line* ℓ *passes through the point where the coins of radii* 1 *and* 2 *touch and the point where the coins of radii* 2 *and* 3 *touch. Find the distance from the centre of the coin of radius* 1 *to the point where* ℓ *meets the line through the centres of the coins of radii* 1 *and* 3.



Let the coins be ω_1, ω_2, ω_3 with centres O_1, O_2, O_3, so $O_1O_3 = 1 + 3 = 4$. The coins touch externally, so the contact points T_1 of ω_1, ω_2 and T_2 of ω_2, ω_3 are internal centres (Theorem 3(2)). By Monge's theorem, line T_1T_2 passes through the external centre P of ω_1 and ω_3.

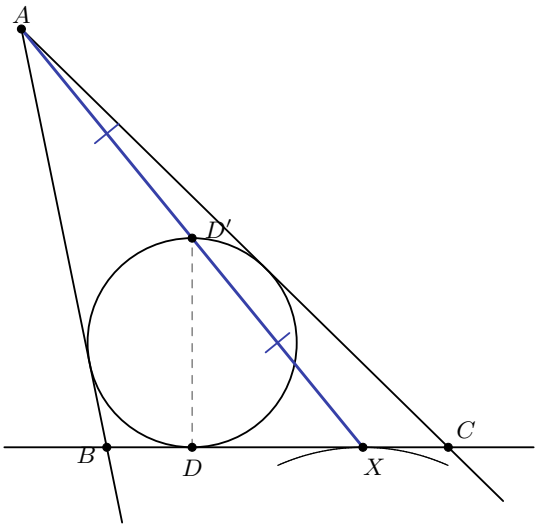
So P lies on line O_1O_3, outside the segment, with $PO_1 : PO_3 = 1 : 3$. So P is on the side of O_1, and $PO_3 = PO_1 + 4$ gives $3\,PO_1 = PO_1 + 4$, that is $PO_1 = 2$.

Answer: 2

Problem 2

The incircle of triangle ABC *touches* BC *at* D, *and* DD' *is a diameter of the incircle. Line* AD' *meets* BC *at* X. *Prove that* D' *is the midpoint of* AX *if and only if* $AB + AC = 3\,BC$.

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By Lemma 5, X is the point where the A-excircle touches BC, and the homothety at A with ratio $r_a/r > 1$ sends D' to X. So D' lies on segment AX with $AD' : AX = r : r_a$, and D' is the midpoint of AX if and only if $r_a = 2r$.

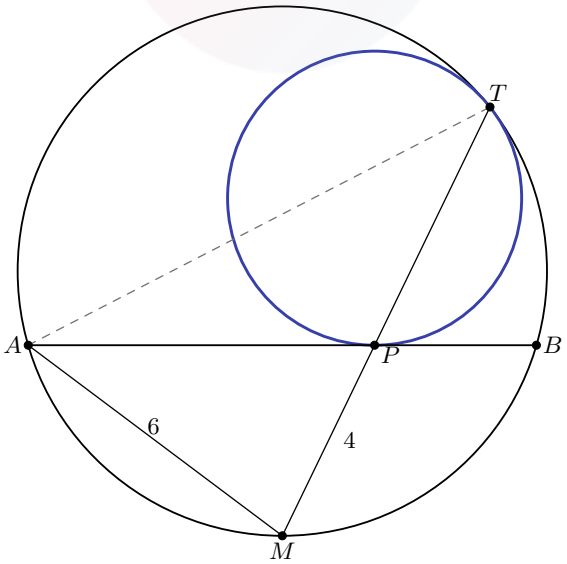
Let Δ be the area and a, b, c the sides. Splitting at I gives $\Delta = rs$. Splitting at I_a gives $\Delta = [ABI_a] + [ACI_a] - [BCI_a] = \frac{1}{2}r_a(c + b - a) = r_a(s - a)$. Hence

$$\frac{r_a}{r} = \frac{s}{s-a},$$

which equals 2 if and only if $s = 2a$, that is $a + b + c = 4a$, that is $b + c = 3a$. ■

Problem 3

Circle ω lies inside circle Γ and touches Γ at T. A chord AB of Γ touches ω at P, and M is the midpoint of the arc AB of Γ not containing T. Given $MA = 6$ and $MP = 4$, find PT.



By the argument at the start of this unit, the homothety at T taking ω to Γ sends P to M, so T, P, M are collinear with P between T and M.

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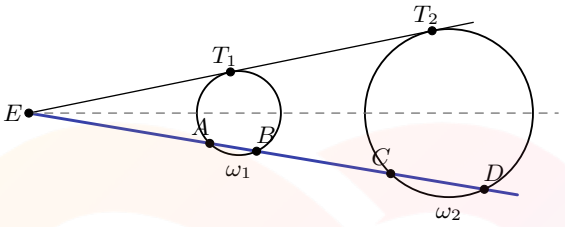
Since $MA = MB$, $\angle MAB = \angle MBA$, and $\angle MBA = \angle MTA$ as both stand on the arc MA. So $\angle MAP = \angle MTA$, and triangles MAP and MTA, which share the angle at M, are similar. Hence $\frac{MA}{MT} = \frac{MP}{MA}$, so $MT = \frac{36}{4} = 9$ and $PT = 9 - 4 = 5$.

In the language of L4-GE2, MA is tangent to circle APT and $MA^2 = MP \cdot MT$ is the power of M.

Answer: 5

Problem 4

Circles ω_1 and ω_2 lie outside each other. A common external tangent touches them at T_1 and T_2 and meets their line of centres at E. A line through E meets ω_1 at A and B and ω_2 at C and D, in the order E, A, B, C, D. Prove that $EA \cdot ED = ET_1 \cdot ET_2$.



By Theorem 3(1), E is the external centre, and the homothety h at E with ratio $k = r_2/r_1$ takes ω_1 to ω_2. As $k > 0$, h sends a point of ray EA at distance t from E to the point of the same ray at distance kt. So h sends A and B to the two points of ω_2 on this ray, keeping their order, so $h(A) = C$ and $h(B) = D$. In the same way the tangent line ET_1 goes to itself and $h(T_1) = T_2$.

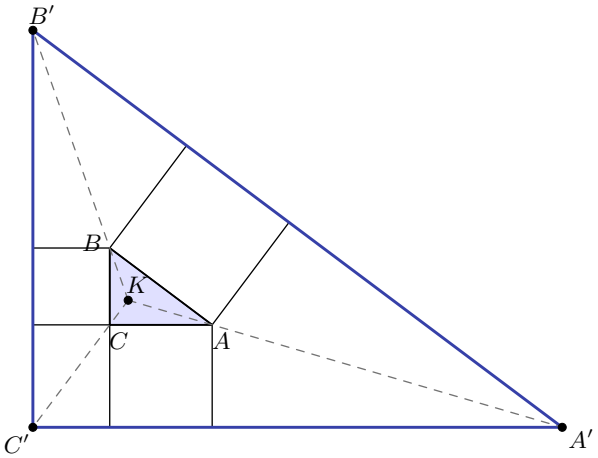
Therefore $ED = k \cdot EB$ and $ET_2 = k \cdot ET_1$. By power of a point, $EA \cdot EB = ET_1^2$. So

$$EA \cdot ED = k \cdot EA \cdot EB = k \cdot ET_1^2 = ET_1 \cdot ET_2. \blacksquare$$

Also $EB \cdot EC = k \cdot EA \cdot EB$ is the same constant, so for two such lines the four points B, C, B', C' are concyclic by the converse of power of a point.

Problem 5

Squares are drawn outwards on the three sides of a right-angled triangle with sides 3, 4 and 5. For each square, the side opposite the triangle is extended to a full line. Find the area of the triangle bounded by these three lines.



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Let the triangle be ABC with $a = BC$, $b = CA$, $c = AB$, area Δ and $\Sigma = a^2 + b^2 + c^2$. The outer side of the square on BC lies on the line parallel to BC at distance a from BC, on the far side from A, and similarly for the other two. So the new triangle has sides parallel to those of ABC, and by Theorem 2 the new triangle should be a homothetic image. We find the centre.

Let $K = \frac{a^2A+b^2B+c^2C}{\Sigma}$. By mass points (L4-GE2, Theorem 8), $[KBC] : [KCA] : [KAB] = a^2 : b^2 : c^2$, so K is inside ABC and its distance to BC is $p_a = \frac{2[KBC]}{a} = \frac{2a\Delta}{\Sigma}$, with $p_b = \frac{2b\Delta}{\Sigma}$ and $p_c = \frac{2c\Delta}{\Sigma}$ similarly. The homothety at K with ratio $k = 1 + \frac{\Sigma}{2\Delta}$ sends line BC to the parallel line on the far side of BC at distance kp_a from K, that is at distance $(k-1)p_a = a$ from BC. This is the line of the outer side of the square on BC, and the same holds for the other two sides.

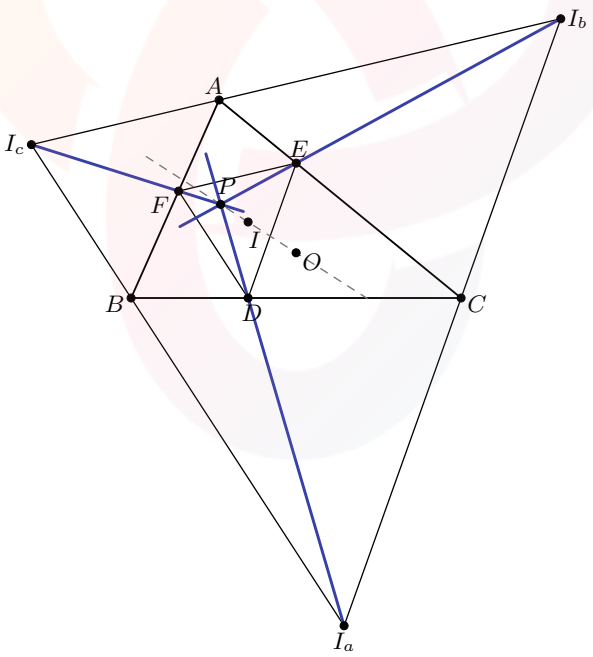
So the new triangle is the image of ABC, with area $k^2\Delta$. Here $\Delta = 6$ and $\Sigma = 50$, so $k = 1 + \frac{50}{12} = \frac{31}{6}$ and the area is $6 \cdot \frac{961}{36} = \frac{961}{6}$.

The point K is the symmedian point of ABC, but nothing about symmedians was needed.

Answer: $\frac{961}{6}$

Problem 6

Let ABC be a triangle that is not equilateral, with incentre I, circumcentre O and excentres I_a, I_b, I_c. The incircle touches BC, CA, AB at D, E, F. Prove that the lines I_aD, I_bE, I_cF are concurrent at a point of line OI.



The internal and external bisectors at A are perpendicular, and I and I_a lie on the internal one and I_b, I_c on the external one. Since $AE = AF$ and $IE = IF$, line AI is the perpendicular bisector of EF, so $EF \perp AI$ and hence $EF \parallel I_bI_c$. Similarly $FD \parallel I_cI_a$ and $DE \parallel I_aI_b$.

Triangle DEF lies inside ABC, which lies inside $I_aI_bI_c$ (its vertices are on the sides), so the two triangles have different areas and are not translates. By Theorem 2 there is a homothety h with $D \mapsto I_a$, $E \mapsto I_b$, $F \mapsto I_c$, and the three lines pass through its centre P. Also h takes the incircle, the circumcircle of DEF, to the circumcircle of $I_aI_bI_c$, so h sends I to that circle's centre V, and P lies on line IV.

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We still need to show that V is on line OI. In triangle $I_aI_bI_c$ the line I_aA is perpendicular to I_bI_c, and similarly at B and C. So A, B, C are the feet of its altitudes and I is its orthocentre. By Theorem 6 its nine-point circle passes through A, B, C, so this circle is the circumcircle of ABC, and its centre O is the midpoint of I and V. As the triangle is not equilateral, $O \neq I$, so V is on line OI, and so is P. ■

The hard step is the last one, seeing ABC as the orthic triangle of $I_aI_bI_c$, so that its circumcircle is a nine-point circle.